

Carlynda Gao

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EDUCATION

University of California, Berkeley

B.A. in Computer Science, Mathematics, & Philosophy

- **Relevant coursework:** Multivariable Calculus, Foundations of Data Science, Linear Algebra & Differential Equations, Discrete Mathematics, Real Analysis, Abstract Linear Algebra, Complex Analysis, Full-Stack Web Development, Structure & Interpretation of Computer Programs, Data Structures & Algorithms, Computer Architecture, Computer Graphics, Optimization Models in Engineering, Intermediate Microeconomics

EXPERIENCE

UC Berkeley Department of Electrical Engineering & Computer Sciences

Research Assistant at the Berkeley Lab for Usable & Experimental Security (BLUES)

Aug. 2024 - May 2025

- Contributed to research focused on improving privacy guidance for app developers of child-directed apps through qualitative coding & thematic analysis of 216 pages of semi-structured interviews with 15 domestic & international developers.
- Compiled domestic & international privacy laws & cybersecurity regulations & utilized full-stack technologies to build & update a guidance web application for child-directed app developers to use as privacy guidance during development.

Cornell University S.C. Johnson College of Business

Research Intern

June 2023 - Dec. 2023

- Authored a 10-page paper on the business operations of the U.S. boba shop industry by tracing the international supply chain, quantifying the economic impact of COVID, & predicting future economic trends through statistical analysis.
- Conducted a 12-page review on East Asian immigration patterns in the 20th century, the historical proliferation of East Asian businesses, & performed case studies on the impact of economic discrimination with qualitative analysis of historical data.

PROJECTS

Ambient GI with Caching | 28sean.github.io/cs184-final-project/final/index.html

C++, C, HTML

- Facilitated the creation of an octree-based ambient global illumination caching system, accelerating indirect lighting computation through caching & interpolation to achieve 15 - 30× speedup over Monte Carlo path tracing for architectural visualization.
- Completed literature reviews of grid-based & octree-based cache algorithms, implemented & optimized cache memory management & interpolation techniques, & conducted scene rendering tests & debugging for grid & octree data structures.

Butterfly | github.com/GottaGetPaid/butterfly-fix

React, Vite, Javascript, HTML, Tailwind CSS, Node.js, Express.js, Next.js, Postman, MongoDB, Spotify Web API

- Collaborated on the creation of Butterfly, a website utilizing MongoDB to allow users to create accounts & profiles to match with other users with similar music taste based on the users' top songs, artists, & genres, tested with Postman.
- Created a recommendations page which uses the Spotify Web API to generate a list of songs, albums, & artists that match the user's algorithmic preferences, with the option to upvote or downvote each suggestion to influence future suggestions.

Personal Website | carlyndaxgao.com

HTML, CSS, Tailwind, Javascript, Three.js, Express.js, Node.js, Next.js, Vercel

- Built a personal website from scratch with full-stack technologies to showcase my projects, interests, skills, & experiences.

SKILLS

- **Languages:** Python, Java, C, C++, Matlab, HTML, CSS, JavaScript, TypeScript, SQL
- **Technologies:** React, Vite, Tailwind, Three.js, Node.js, Express.js, Next.js, MongoDB, NumPy, Pandas, Matplotlib, Seaborn, Scipy, Jupyter, Colab, Git, Github, Vercel, Postman, Microsoft Office, Figma, Blender